Two-Photon Physics at a Lepton-Proton Collider at CERN

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Abstract

High-energy photon-photon scattering is a complementary component of the physics programme at future CERN lepton-proton facilities. We examine the equivalent-photon description of elastic and proton-dissociative two-photon reactions and discuss their prospects at the Large Hadron-electron Collider (LHeC) and a proposed muon-proton collider (LHμC). The photon luminosity determines the rate of a hard two-photon subprocess. Two illustrative precision benchmarks are emphasized: tau-lepton pair production as a probe of electromagnetic dipole interactions, and W-boson pair production as a test of the electroweak quartic gauge sector. We distinguish previously studied collider benchmarks from newer, higher-energy muon-beam proposals and discuss how high-invariant-mass distributions may amplify beyond-Standard-Model effects. Loop-induced neutral gauge-boson production, top pairs, charged electroweak states and light-by-light scattering provide additional opportunities. The quantitative interpretation of anomalous-coupling effects requires consistent photon-emission selections, effective-field-theory validity checks and detector-level studies.

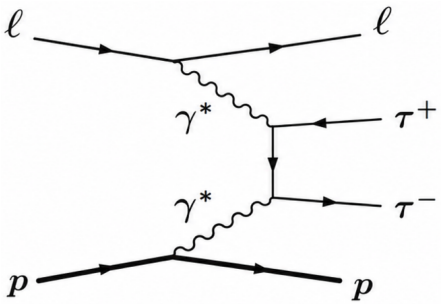
1 Introduction

Photon-photon fusion offers a particularly transparent route to the production of electrically charged and electroweakly coupled final states. A relativistic lepton and proton radiate photons, which subsequently undergo a hard collision, $\gamma\gamma \to X$. The radiating particles may remain intact, giving a fully exclusive topology, or the proton may dissociate into a low- or high-mass hadronic system. This separation of electromagnetic emission and the short-distance subprocess underlies the equivalent photon approximation (EPA) [1, 2].

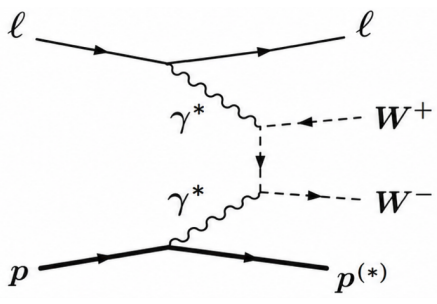
The proposed Large Hadron-electron Collider (LHeC) combines a lepton beam of order 50 GeV with the 7 TeV LHC proton beam, yielding $\sqrt{s_{ep}} \simeq 1.2\,\text{TeV}$ [3, 4]. An electron-proton environment is especially attractive for exclusive studies because it has one rather than two colored incoming beams and is expected to have much less event pileup than high-luminosity proton-proton collisions. Neither advantage removes detector inefficiencies, proton dissociation or all soft-interaction uncertainties, but both simplify the experimental interpretation of photon-fusion signatures [5, 6].

A complementary LHμC concept uses energetic positive muons against LHC proton beams. The recent proposal of Ref. [7] emphasizes 1 and 2 TeV muon beams, leading to $\sqrt{s_{\mu p}} \simeq 5.3$ and 7.5 TeV, respectively, with a projected average luminosity approaching $10^{34}\,\text{cm}^{-2}\text{s}^{-1}$. These are machine scenarios: published photon-fusion predictions must be associated with the particular energy and photon selections actually used in each calculation.

We outline the EPA and the energy dependence of its luminosity function, and focus on two physics cases that connect Standard-Model (SM) cross sections with higher-dimensional interactions: $\gamma\gamma \to \tau^+\tau^-$ and $\gamma\gamma \to W^+W^-$. Other charged and neutral final states complete a broader programme in which a high-energy photon spectrum, clean event topology and differential observables are complementary assets.



(a) $\gamma\gamma \to \tau^+\tau^-$



(b) $\gamma\gamma \to W^+W^-$

Figure 1: Representative Standard Model Feynman diagrams for photon-induced $\tau^+\tau^-$ (left) and W^+W^- (right) production in lepton–proton collisions, $\ell p \to \ell X p^{(*)}$, with $\ell = e, \mu^+$.

2 Theoretical framework for two-photon interactions

2.1 Equivalent photon approximation

For $\ell = e^-$ or μ^+, the generic reaction is $\ell p \to \ell + X + p^{(*)}$. In the quasi-real limit the cross section can be written as a convolution of photon spectra and the elementary photon-photon cross section,

$$\sigma_{\ell p \to \ell X p^{(*)}} = \int_0^1 dy_\ell \int_0^1 dy_p \, \Phi_\ell(y_\ell)\Phi_p(y_p) \, \hat{\sigma}_{\gamma\gamma\to X}(W_{\gamma\gamma}), \tag{1}$$

where y_i is the fractional energy carried by the photon and $\Phi_i(y_i) = dN_{\gamma/i}/dy_i$. Here $W_{\gamma\gamma}$ denotes the diphoton invariant mass, with $W^2_{\gamma\gamma} \simeq y_\ell y_p s_{\ell p}$ for collinear, quasi-real emission. Its natural generalization is

$$\begin{aligned} \sigma_{\ell p} &= \int_{W_{\rm thr}}^{\sqrt{s_{\ell p}}} dW \, S_{\gamma\gamma}(W) \, \hat{\sigma}_{\gamma\gamma\to X}(W), \\ S_{\gamma\gamma}(W) &= \frac{2W}{s_{\ell p}} \int_{W^2/s_{\ell p}}^1 \frac{dy_\ell}{y_\ell} \, \Phi_\ell(y_\ell)\Phi_p\!\left(\frac{W^2}{s_{\ell p} y_\ell}\right). \end{aligned} \tag{2}$$

Thus $S_{\gamma\gamma}$ has dimensions of inverse energy, and its integral over W gives a dimensionless effective two-photon luminosity fraction. Equation (2) is a collinear EPA identity, not an exact statement at arbitrarily high photon virtuality.

2.2 Equivalent-photon spectra and photon virtuality

For a point-like lepton of mass m_ℓ, an illustrative virtuality-integrated EPA spectrum is [1]

$$\begin{aligned} \Phi_\ell(y) &= \frac{\alpha}{2\pi}\left[\frac{1+(1-y)^2}{y} \ln\frac{Q^2_{\rm max}}{Q^2_{\rm min}} - \frac{2(1-y)}{y}\left(1 - \frac{Q^2_{\rm min}}{Q^2_{\rm max}}\right)\right], \\ Q^2_{\rm min} &\simeq \frac{m_\ell^2 y^2}{1-y}. \end{aligned} \tag{3}$$

for $Q^2_{\rm min} < Q^2_{\rm max}$, with a vanishing flux otherwise. The muon mass increases $Q^2_{\rm min}$ relative to an electron at fixed y: the gain from a muon collider therefore comes from its accessible beam energy and usable phase space, not from a universally larger flux at fixed energy. For an elastic proton, an often used representation is

$$\Phi_p^{\rm el}(y) = \frac{\alpha}{\pi y}\int_{Q^2_{\rm min}}^{Q^2_{\rm max}} \frac{dQ^2}{Q^2}\left[(1-y)\left(1-\frac{Q^2_{\rm min}}{Q^2}\right)F_E(Q^2) + \frac{y^2}{2}F_M(Q^2)\right], \tag{4}$$

where $F_E = (4m_p^2 G_E^2 + Q^2 G_M^2)/(4m_p^2 + Q^2)$ and $F_M = G_M^2$ involve the Sachs electromagnetic form factors [2]. Here $Q^2 = -q^2 > 0$ denotes the photon virtuality, and the integration bounds must be matched to the tagging or acceptance cuts. For the proton-dissociative channel $p \to \gamma^* + N^*$, the photon spectrum can instead be expressed through measured inclusive electromagnetic structure functions $F_2(x, Q^2)$ and $F_L(x, Q^2)$ [8, 9]. With M_N the dissociation mass, $x = Q_p^2/(M_N^2 - m_p^2 + Q_p^2)$, so that

$$M_N^2 - m_p^2 = Q_p^2\left(\frac{1}{x} - 1\right). \tag{5}$$

An expanded dissociation phase space can raise the predicted rate, but it changes the final-state topology and increases sensitivity to the structure-function treatment. In particular, an inclusive photon PDF and a proton-tagged elastic flux should not be regarded as interchangeable inputs.

2.3 Validity of the approximation

For $Q_\ell^2, Q_p^2 \ll W_{\gamma\gamma}^2$, transverse quasi-real photons dominate, and Eqs. (1)-(2) provide a well-motivated factorized description of photon-induced lepton-proton interactions. The commonly used high-energy relation $W_{\gamma\gamma}^2 \simeq y_\ell y_p s_{\ell p}$, however, neglects kinematic corrections associated with photon virtualities and proton dissociation. Within the kinematic conventions adopted in the present EPA treatment, the corrected expressions for the two-photon invariant mass are

$$\begin{aligned} W_{\gamma\gamma,\rm el}^2 &= -Q_\ell^2 + y_\ell y_p s_{\ell p}, \\ W_{\gamma\gamma,\rm inel}^2 &= -Q_\ell^2 - Q_p^2 + y_\ell y_p s_{\ell p} \\ &\quad + \frac{Q_\ell^2}{s_{\ell p}}\left(Q_p^2 + M_N^2 - m_p^2\right). \end{aligned} \tag{6}$$

Here Q_ℓ^2 and Q_p^2 denote the virtualities of the photons emitted by the lepton and proton, respectively, y_ℓ and y_p are their longitudinal energy fractions, and $s_{\ell p}$ is the squared lepton-proton centre-of-mass energy. The mass M_N characterizes the proton-dissociative hadronic system, while m_p denotes the proton mass. The elastic expression explicitly retains the lepton-side virtuality correction, whereas the proton-dissociative expression also depends on Q_p^2 and M_N.

Larger photon virtualities can modify the $\gamma^*\gamma^*$ matrix elements, accessible phase space, and contributions from longitudinal photon polarizations. Consequently, full virtuality-dependent calculations become preferable when $Q_{\ell,p}^2$ is not negligible compared with $W_{\gamma\gamma}^2$. Comparisons with matrix-element generators can yield percent-level agreement for elastic production when consistent kinematic definitions and phase-space selections are employed, while proton-dissociative predictions are more sensitive to the allowed M_N and Q_p^2 ranges. Accurate reconstruction of $W_{\gamma\gamma}$ and explicit control of the photon-virtuality assumptions are therefore essential for reliable comparisons between the LHeC and LHμC configurations.

3 Photon-photon luminosity at the LHeC and LHμC

Table 1 summarizes representative beam configurations. The relation $s_{\ell p} \simeq 4E_\ell E_p$ is sufficient for the displayed high-energy values. The 0.5 TeV muon scenario was used in

Table 1: Representative LHeC and LHμC benchmark configurations, assuming a 7 TeV proton beam [7].

Configuration	E_ℓ [GeV]	$\sqrt{s_{\ell p}}$ [TeV]
LHeC ($e^- p$)	20	0.75
LHeC ($e^- p$)	50	1.18
LHμC (option 1)	500	3.74
LHμC (baseline)	1000	5.29
LHμC (option 2)	2000	7.48

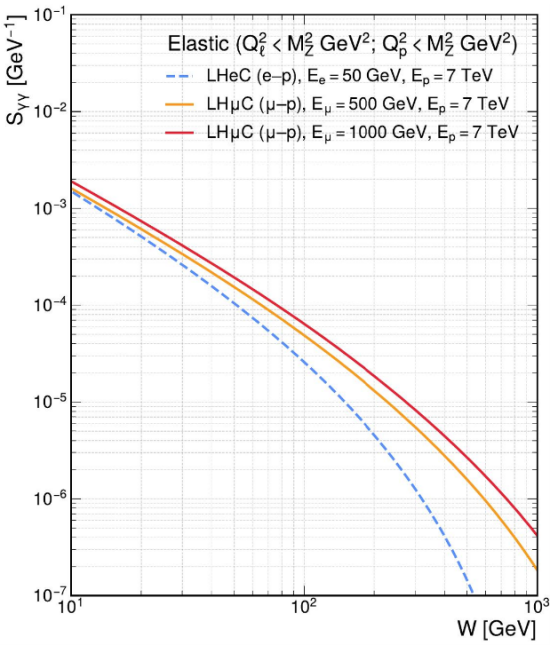


Figure 2: Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}(W)$ as a function of the two-photon invariant mass W at the LHeC ($E_e = 50$ GeV) and LHμC ($E_\mu = 500$ and 1000 GeV), assuming a 7 TeV proton beam. The higher-energy muon–proton configurations provide an enhanced photon–photon luminosity in the large-W region, extending the kinematic reach for heavy final states.

earlier phenomenological studies, while the 1 and 2 TeV configurations are highlighted in the newer machine proposal [7]. A higher $\sqrt{s_{\ell p}}$ permits a harder photon–photon spectrum; however, neither the shape nor normalization of $S_{\gamma\gamma}$ can be inferred from $\sqrt{s}$ alone.

As illustrated in Fig. 2, the elastic photon-photon luminosity spectrum decreases rapidly with increasing invariant mass W. However, the higher lepton-beam energies available at the LHμC result in a harder spectrum than at the LHeC. Already for $E_\mu = 500$ GeV, the luminosity is enhanced in the high-W region, while the 1 TeV muon option extends this advantage further. This enhancement is particularly relevant for the production of heavy final states and for probing energy-dependent contributions from anomalous interactions. In the region of larger W, the increase in the kinematic energy scale compensates for the steeper suppression associated with high photon momentum fractions. The quantitative ratio is nevertheless dependent on the allowed Q^2 range, proton emission mode and W selection, and the plotted example must not be used as an experimentally validated luminosity forecast.

Table 2: Elastic photon-induced production cross sections for $\tau^+\tau^-$ and W^+W^- pairs at the LHeC and LHμC benchmark configurations. The quoted values correspond to integrated $\ell p \to \ell X p$ production cross sections, with $X = \tau^+\tau^-$ or W^+W^-, before decay branching fractions and detector effects.

Collider	$\sqrt{s_{\ell p}}$ [TeV]	$\sigma^{\rm el}_{\tau\tau}$ [pb]	$\sigma^{\rm el}_{WW}$ [pb]
LHeC	1.2	50.40	0.037
LHμC	3.7	58.5	0.166
LHμC	5.3	70.3	0.258

4 Tau-pair production and electromagnetic moments

4.1 Standard-Model production

The tree-level process $\gamma\gamma \to \tau^+\tau^-$ is governed by QED t- and u-channel tau exchange. The relatively low production threshold makes this channel valuable for photon-luminosity calibration, while the large accessible invariant mass enables a test of the $\gamma\tau\tau$ vertex well above the tau mass. The first observation of photon-induced tau-pair production in pp collisions, together with bounds on tau electromagnetic moments, has already demonstrated the utility of this mechanism [10].

The collaborative benchmark values collected in Table 2 illustrate a modest increase in the total elastic tau-pair rate as the muon beam energy is raised. Since the underlying acceptance and virtuality definitions could not be recovered here in a machine-readable form, these numbers are treated as *indicative reference inputs*, not independently reproduced predictions; they are intentionally not combined with the separate illustrative flux calculation of Fig. 2. The key physical observation is more robust than any single integral: more energetic photon-photon collisions extend the measurable $M_{\tau\tau}$ distribution beyond a low-mass dominated inclusive sample.

4.2 Dipole interactions and their EFT interpretation

A convenient on-shell description of non-standard tau electromagnetic interactions uses

$$\mathcal{L}^{\rm dip}_{\tau\tau\gamma} = \frac{e\,\delta a_\tau}{4m_\tau}\,\bar\tau\sigma^{\mu\nu}\tau F_{\mu\nu} - \frac{i\,d_\tau}{2}\,\bar\tau\sigma^{\mu\nu}\gamma_5\tau F_{\mu\nu}, \tag{7}$$

where δa_τ denotes the deviation from the SM magnetic moment and d_τ is the electric dipole moment; the sign conventions are conventional and must be kept fixed when matching amplitudes. The magnetic term is CP even, whereas the electric term violates CP. At scales above electroweak symmetry breaking, the corresponding new-physics interactions may be embedded in gauge-invariant dimension-six SMEFT operators,

$$\mathcal{O}^{33}_{eB} = (\bar L_3\sigma^{\mu\nu}\tau_R)HB_{\mu\nu}, \qquad \mathcal{O}^{33}_{eW} = (\bar L_3\sigma^{\mu\nu}\tau_R)\sigma^I HW^I_{\mu\nu}, \tag{8}$$

with $\Delta\mathcal{L} = (C^{33}_{eB}\mathcal{O}^{33}_{eB} + C^{33}_{eW}\mathcal{O}^{33}_{eW})/\Lambda^2 + \text{h.c.}$. The physical photon coupling depends on the weak mixing angle and on the real and imaginary parts of the Wilson-coefficient combination. The exact numerical matching should follow the normalization chosen for the SU(2) generators and the dipole vertex.

4.3 High-mass distributions and sensitivity

The amplitude may be organized schematically as $\mathcal{M} = \mathcal{M}_{\rm SM} + \delta a_\tau\mathcal{M}_a + d_\tau\mathcal{M}_d$. Consequently a differential distribution contains interference and higher-order dipole terms, while the CP-odd interference may vanish in sufficiently inclusive CP-even observables. The behavior of $d\sigma/dM_{\tau\tau}$, rather than only the integrated cross section, can

therefore improve the separation of anomalous scenarios from the SM. This enhancement must be weighed against the rapid decrease in photon luminosity and the efficiency of reconstructing visible tau decays with escaping neutrinos.

Earlier exploratory plots used $\delta a_\tau = 0.008$, whereas an updated study considered $\delta a_\tau = 5\times10^{-4}$. Because the original numerical files and curve definitions are not available in the present portable package, we do not reproduce or relabel the corresponding distributions. A genuine exclusion analysis would need binned event yields, backgrounds, systematic uncertainties, correlations and an explicit treatment of the dipole operator's range of validity. It is therefore more accurate here to identify the high-$M_{\tau\tau}$ region as a promising *observable* than to advertise a new confidence-level reach.

5 W-boson pairs and anomalous quartic gauge couplings

5.1 Standard-Model $\gamma\gamma \to W^+W^-$

At leading order $\gamma\gamma \to W^+W^-$ proceeds through charged-W exchange in the t and u channels, together with the four-point $\gamma\gamma W^+W^-$ vertex. Gauge invariance ties these amplitudes together. Unlike massive charged scalar or fermion-pair production, the unpolarized high-energy vector-pair subprocess retains a sizable rate, making it an unusually productive probe of the electroweak gauge sector [6]. Its threshold is $W_{\rm thr} = 2m_W$, substantially higher than for tau pairs.

For the elastic benchmarks summarized in Table 2, the W^+W^- production cross section at the LHμC exceeds the corresponding LHeC reference value by factors of approximately 4.5 and 7.0 at $\sqrt{s} = 3.7$ and 5.3 TeV, respectively. Assuming an illustrative integrated luminosity of $250\,\mathrm{fb}^{-1}$ for each configuration, the corresponding numbers of produced W-boson pairs are 9.25×10^3 at the LHeC, compared with 4.15×10^4 and 6.45×10^4 at the LHμC for $\sqrt{s} = 3.7$ and 5.3 TeV, respectively, before accounting for branching fractions, detector acceptance, and reconstruction efficiencies.

These results illustrate the substantial enhancement of photon-induced electroweak production at the LHμC, driven by its higher collision energy and harder photon-photon luminosity spectrum. The increased event rates, together with the extended high-invariant-mass reach, provide promising opportunities for more precise measurements and improved sensitivity to anomalous electroweak interactions. Quantitative sensitivity projections, however, require consistent event selections, background estimates, and detector-level reconstruction.

5.2 Dimension-eight quartic interactions

Anomalous quartic gauge couplings (aQGCs) are conveniently parameterized in a linearly realized electroweak EFT using higher-dimensional operators. A generic representation is

$$\mathcal{L}_{\rm EFT} = \mathcal{L}_{\rm SM} + \sum_i \frac{f_i}{\Lambda^4}\mathcal{O}_i^{(8)} + \cdots, \qquad \sigma = \sigma_{\rm SM} + \sum_i \frac{f_i}{\Lambda^4}\sigma_{{\rm int},i} + \sum_{ij} \frac{f_i f_j}{\Lambda^8}\sigma_{{\rm quad},ij}. \quad (9)$$

The commonly used S, M and T operator families organize contributions involving covariant Higgs derivatives and electroweak field strengths [11, 12]. For example, in one standard convention $\mathcal{O}_{M2} = (B_{\mu\nu}B^{\mu\nu})[(D_\beta H)^\dagger D^\beta H]$. A benchmark such as $f_{M2}/\Lambda^4 = 1\,\mathrm{TeV}^{-4}$ changes the amplitude, but is *not* by itself an experimental constraint. Equation (9) also makes clear that large positive high-mass deviations can be dominated by the quadratic EFT term, potentially in a regime where the truncation becomes unreliable.

5.3 Comparison and EFT domain of validity

The principal sensitivity of $\gamma\gamma \to WW$ to anomalous contributions occurs in high-M_{WW} bins, where increased hard-scattering energy may offset declining photon luminosity. A collider-to-collider comparison must hold the EFT coefficients, final-state acceptance, photon virtuality cuts and proton-emission topology fixed. In particular, the original LHμC aQGC distribution used a 5.3 TeV collider configuration, while the elastic SM cross-section number just quoted corresponds to 3.7 TeV; one must not mix them into a single numerical limit. Furthermore, an LHC proton–proton comparison is affected by proton survival probabilities, pileup and tagging conditions that have no simple counterpart in ℓp scattering.

An experimentally interpretable analysis would report the selected reconstructed final state, detector response, background template, correlated uncertainties and treatment of parameter degeneracies. It would also specify an energy domain such that the operator expansion remains controlled. A visible ratio $[d\sigma/dM_{WW}]_{\mathrm{EFT}}/[d\sigma/dM_{WW}]_{\mathrm{SM}}$ signals a potentially discriminating observable; it is not equivalent to a detector-level 95% confidence interval. This distinction is central when comparing upgraded LHμC energies with earlier baseline studies.

6 Additional Standard-Model and BSM opportunities

Beyond tau and W pairs, several photon-fusion processes become particularly interesting as the accessible diphoton invariant mass increases. The neutral final state $\gamma\gamma \to ZZ$ is loop-induced in the SM, making its high-mass spectrum a promising probe of anomalous neutral quartic gauge interactions. The process $\gamma\gamma \to t\bar{t}$ provides a complementary electromagnetic production mechanism for the heaviest SM fermion and offers sensitivity to possible deviations in its photon couplings. With a production threshold near $2m_t$, its rate depends strongly on the high-energy tail of $S_{\gamma\gamma}$ and must be distinguished from the substantially larger γg top-quark photoproduction channel.

Charged electroweak states provide further opportunities for probing physics beyond the SM. Higgsino-like chargino and charged-slepton pairs can be produced through $\gamma\gamma \to \widetilde{H}^+\widetilde{H}^-$ and $\gamma\gamma \to \widetilde{\ell}^+\widetilde{\ell}^-$, respectively. Their leading electromagnetic interactions are determined by the electric charge, while their spin and mass govern the threshold behaviour and angular distributions. Compressed mass spectra remain experimentally challenging because the visible decay products may be soft and the missing-momentum signatures require accurate event reconstruction. Existing LHC searches already constrain parts of this parameter space, making photon-induced production at lepton–hadron colliders a complementary approach.

Finally, the loop-induced processes $\gamma\gamma \to \gamma\gamma$ and $\gamma\gamma \to H$ provide additional probes of charged virtual particles and potential BSM contributions. Their relatively small production rates emphasize the importance of controlling photon spectra, exclusive-event selections, and backgrounds. Compared with the LHeC, the higher centre-of-mass energies available at the LHμC lead to a harder photon–photon luminosity spectrum, enhancing the accessible high-W region. This advantage is particularly relevant for heavy final states, where production thresholds strongly suppress the LHeC rates, and for anomalous interactions whose effects grow with energy. Consequently, the LHμC offers the potential for substantially improved mass reach and enhanced sensitivity to electroweak and BSM interactions, subject to realistic luminosity, acceptance, and background assumptions. These additional channels significantly broaden the photon-induced physics programme while complementing the precision studies of tau and W-boson pairs.

7 Experimental considerations and future prospects

A reliable photon-fusion analysis must simultaneously account for the outgoing-lepton acceptance, central-system reconstruction, and proton final state. Elastic channels may exploit forward proton tagging, large rapidity gaps, small transverse-momentum imbalance, and acoplanarity, whereas proton-dissociative channels require either explicit hadronic reconstruction or a carefully modelled inclusive treatment. Tau decays introduce neutrinos, while W-boson reconstruction involves branching fractions, combinatorial ambiguities, and missing-energy effects. Searches for new charged particles may additionally be limited by detector acceptance and reconstruction efficiencies. These experimental effects are not fully captured by generator-level EPA calculations.

The machine-detector interface for the LHμC remains under development [7]. Quantitative sensitivity studies must therefore account for muon-beam-induced backgrounds, beam optics, forward-tagging acceptance, and realistic integrated luminosities. Future investigations should combine validated photon fluxes with full matrix-element calculations where appropriate, detector simulations, and statistical analyses consistent with EFT validity requirements. The extended calculations at $\sqrt{s} = 3.7$, 5.3, and 7.5 TeV enable a systematic comparison of the LHμC physics potential across different beam-energy configurations. In particular, the enhanced high-invariant-mass photon-photon luminosity provides greater kinematic reach for heavy charged states and offers promising opportunities for improved sensitivity to anomalous electroweak interactions relative to the LHeC.

8 Summary and conclusions

High-energy photon fusion offers a complementary electroweak and BSM physics programme at future CERN lepton-proton colliders. The EPA organizes the connection between beam-dependent photon fluxes and the hard $\gamma\gamma$ subprocess, while elastic and proton-dissociative emission must remain distinct. Compared with the LHeC, the higher beam energies attainable at the LHμC provide a harder $\gamma\gamma$ luminosity spectrum and substantially enhance the accessible high-invariant-mass region, offering improved sensitivity reach to anomalous electroweak interactions and heavy charged states. In particular, tau-pair spectra and W-pair production furnish promising probes of anomalous electromagnetic dipole moments and quartic gauge couplings, respectively, with potentially enhanced sensitivity at the LHμC. Consistent kinematic selections, reliable photon-emission models, detector response, and EFT validity checks remain indispensable for establishing quantitative sensitivity projections.

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